ACS ES&T Water

Open Access

This article is licensed under CC-BY-NC-ND 4.0

pubs.acs.org/estwater

Article

Spatial Trends and Health Risks of Per- and Polyfluoroalkyl Substances in San Francisco Bay Fish from 2009 to 2019

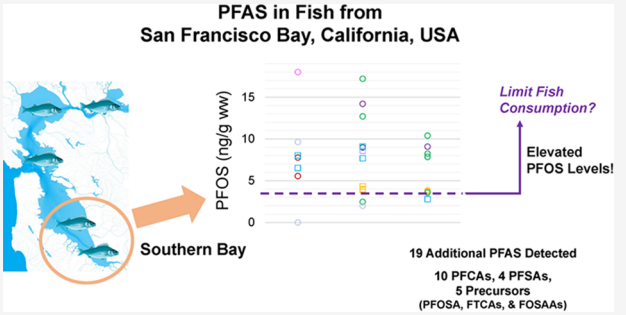
Miguel A. Méndez, Jay Davis, Ezra L. Miller, Richard Grace, and Rebecca Sutton*

Cite This: ACS EST Water 2025, 5, 2903−2913

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Consumption of contaminated food, especially seafood, is a key source of human perfluoroalkyl and polyfluoroalkyl substances (PFAS) exposure. Here, we examine the occurrence of PFAS in shiner surfperch (*Cymatogaster aggregata*), striped bass (*Morone saxatilis*), white croaker (*Genyonemus lineatus*), and seven other sport fish from San Francisco Bay, California, U.S. over a decade of monitoring to assess the potential risks from dietary exposures. In fish collected in 2009, 2014, and 2019, perfluorooctanesulfonic acid (PFOS) was predominantly detected at levels exceeding PFOS consumption advisory thresholds established in other U.S. states. The southern regions of San Francisco Bay have been especially impacted, with over 80% of samples above the strictest U.S. PFOS consumption advisory threshold set in Massachusetts (for one eight-ounce serving a week) at 3.5 ng/g (detected range: 2.0−18 ng/g ww) compared to only 8% in the other subembayments (detected range: 0.59−8.5 ng/g ww). An additional 19 PFAS were detected, with particularly elevated levels of 7:3 fluorotelomer carboxylic acid (7:3 FTCA) found in several species and sites, representing the first observations in marine fish globally. These findings indicate the need to consider a wider range of PFAS in assessing dietary exposure risks and environmental impacts.



KEYWORDS: *PFAS, marine wildlife, spatial variation, dietary exposure, consumption advisories*

INTRODUCTION

Identified by their carbon−fluorine bonds, PFAS are an extensive class of thousands of anthropogenic chemicals with broad uses across consumer, commercial, and industrial products.[1−3] The unique chemical and physical properties of PFAS, especially their extreme stability, surfactant functionality, and oil and water repellency, have allowed for a plethora of applications. This, and their high-volume production, has led to widespread detection in environmental matrices, wildlife, and humans across the globe.[4−6]

Historically, the PFAS produced and found at the highest volumes were PFOS and perfluorooctanoic acid (PFOA), part of subclasses of PFAS known as perfluorosulfonic acids (PFSAs) and perfluorocarboxylic acids (PFCAs), respectively. Both are extremely persistent in the environment, bioaccumulative in terrestrial and aquatic organisms and cause various toxic effects in humans.[7] Several studies have also shown the biomagnification of PFOS and several PFCAs in aquatic ecosystems, with studies finding some of the highest levels of PFOS in tissue from apex predators such as polar bears, whales, and seals.[8−14]

The U.S. phased out production of PFOS, PFOA, and related substances in 2002 and 2015, respectively, while the international Stockholm Convention listed PFOS for restriction in 2009 and PFOA for elimination in 2019. Other PFAS quickly replaced these compounds, including short-chain compounds (PFSAs with ≤6 fluorinated carbons; PFCAs ≤ 7 fluorinated carbons), perfluorinated alkyl ethers, and polyfluorinated substances.[15−17] In general, much less is known about these emerging PFAS with their toxicity and persistence suspected to be of concern. Many of these replacements, known as PFAS precursors, transform into long-chain PFAS in the environment. In 2022, perfluorohexanesulfonic acid (PFHxS) was added to the Stockholm Convention to eliminate its use. Studies are mixed on the bioaccumulation of these replacements, with several PFAS precursors showing a higher bioaccumulation compared to PFOS and long-chain compounds and short-chain PFAS showing lower bioaccumulation potential.[6,11,18] The mobility of these new alternatives in the environment may lead to continuous and lasting exposure on a broader spatial scale.[15,19]

One of the primary routes of PFAS exposure in humans is by consuming contaminated food, with seafood identified as an

Received: October 18, 2024
Revised: March 28, 2025
Accepted: April 8, 2025
Published: May 20, 2025



© 2025 The Authors. Published by American Chemical Society

important dietary source.[20−23] To minimize PFAS exposure risk, 16 U.S. states have developed consumption advisories to advise consumers of recreationally caught fish, generally focused on PFOS as the most prevalent compound posing the greatest known risk.[24−28] Consumption advisories or establishment of criteria for impairment of a water body based on fish PFAS levels is often a risk-assessment-based process specific to individual water bodies and resident fish species. California, the U.S. state with the highest population, is in the process of developing consumption advisory thresholds for some PFAS. The California Office of Environmental Health Hazard Assessment (OEHHA) is reviewing the latest literature and California fish tissue data to determine appropriate Advisory Tissue Levels (ATLs), used in developing consumption advisories.

Beginning in 2009, PFAS have been included in regular sport fish contaminant monitoring in San Francisco Bay, CA, a large urban estuary surrounded by almost eight million people (Figure 1). Sport fish monitoring is conducted by the Regional

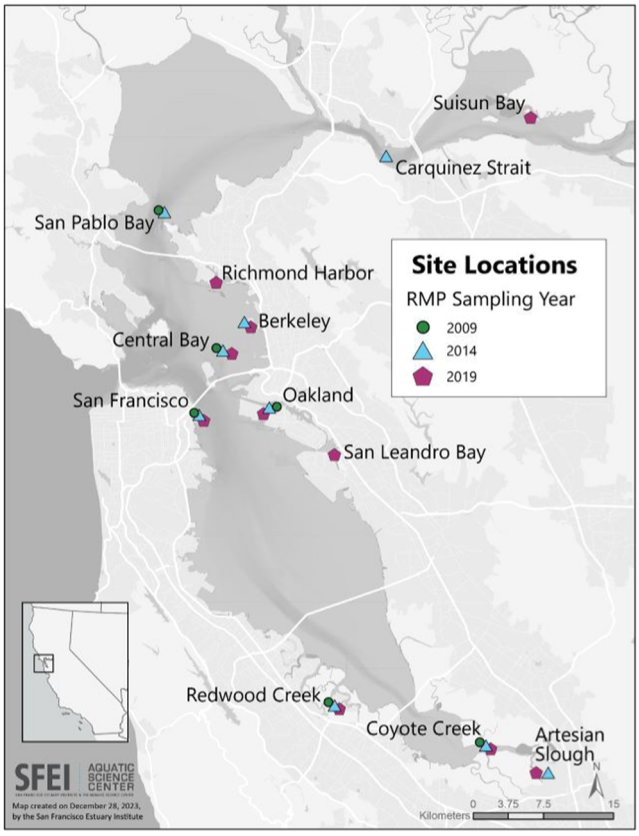


Figure 1. Fish sampling locations in San Francisco Bay in 2009, 2014, and 2019.

Monitoring Program for Water Quality in San Francisco Bay (RMP) and other periodic PFAS monitoring efforts in other biota, sediment, surface water, stormwater, and wastewater. Notably, long-term monitoring of double-crested cormorant (*Phalacrocorax auritus*) eggs and serum from Pacific harbor seals (*Phoca vitulina richardii*) has indicated decreasing levels of PFOS from 2006 to 2016, especially in southern portions of San Francisco Bay, where the highest levels have been consistently detected.[29−31] Such declines are hypothesized to result from the nationwide phase-out of PFOS.

The primary goal of this study was to better understand the occurrence of PFAS in San Francisco Bay sport fish over three sampling events in 2009, 2014, and 2019 and to evaluate potential human exposure relating to fish consumption. Of the ten fish species commonly caught by recreational fishers, 52 samples were collected and analyzed from 2009 to 2019. An additional 55 samples encompassing all three monitoring efforts were archived and later analyzed in 2022 using recently validated EPA Method 1633, including the largest suite of PFAS (40) analytes to date. The greater number of samples and the breadth of monitored PFAS provide a more robust data set for evaluating spatial trends. Levels of PFAS in fish were compared with available consumption advisory thresholds to assess the potential for human health risks. The findings from this study are intended to guide future monitoring and inform management actions to protect aquatic life and human health.

MATERIALS AND METHODS

Sample Collection. Fish were collected at up to 12 stations throughout the San Francisco Bay in three monitoring efforts in 2009, 2014, and 2019 (Figure 1). Five stations are popular recreational fishing locations: three (Oakland, San Francisco, and Redwood Creek) have been continuously sampled during each monitoring effort, and the two other stations (Berkeley and San Pablo Bay) were sampled during only two of the efforts (Figure 1). The Central Bay station has been sampled during each effort to target species found primarily in deeper waters. The remaining six stations (Artesian Slough, Carquinez Strait, Coyote Creek, Richmond Harbor, San Leandro Bay, and Suisun Bay) represent areas of interest to more fully understand contaminant concentrations across San Francisco Bay, with some in close proximity to known PFAS pathways, including wastewater effluent outflows and stormwater runoff. In particular, the Artesian Slough station is directly downstream of the treated effluent discharge from the San José-Santa Clara Regional Wastewater Facility (average flow of 110 million gallons per day), one of the largest advanced wastewater treatment facilities in the western United States. This discharge also influences the Coyote Creek station. These stations were typically sampled less frequently, except for Coyote Creek, which was sampled in all years (Figure 1).

Across sample years, a diverse set of fish species were selected using the following criteria: (1) popular for consumption, (2) sensitive indicators of contamination, (3) widely distributed, (4) representative of different exposure pathways (i.e., benthic and pelagic species), and (5) inclusion in past monitoring efforts. Core species have been consistently collected since 2009 and include shiner surfperch (*Cymatogaster aggregata*), striped bass (*Morone saxatilis*), white croaker (*Genyonemus lineatus*), and white sturgeon (*Acipenser transmontanus*). Additional species were intermittently collected across monitoring years, including California halibut (*Paralichthys californicus*), common carp (*Cyprinus carpio*), largemouth bass (*Micropterus salmoides*), leopard shark (*Triakis semifasciata*), northern anchovy (*Engraulis mordax*), and Pacific staghorn sculpin (*Leptocottus armatus*). Shiner surfperch and Pacific staghorn sculpin are thought to have fairly high site fidelity with home ranges of 2 km or less, while the other species spend less time in San Francisco Bay and can migrate to the Pacific Ocean and freshwaters of the upstream Sacramento-San Joaquin Delta region.[31] Fish were caught by the Marine Pollution Studies Lab at Moss Landing Marine

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

Laboratories (MPSL-MLML) using beach seines, fyke nets, gill nets, hook and line, and otter trawls. Samples were stored frozen at −20 °C before processing by the MPSL-MLML within a month.

Sample composites were prepared by species, generally using skinless fillet muscle tissue, consistent with the OEHHA guidelines to "eat only the skinless fillet of fish".[32] However, small species (e.g., shiner surfperch, northern anchovy, and Pacific staghorn sculpin) were processed whole, with the head, tail, and viscera removed. White croaker were typically prepared as skinless fillets, except in 2014 when they were processed as whole fish. Composites were analyzed to obtain representative data in a cost-effective manner—a common practice recommended as part of U.S. Environmental Protection Agency guidance.[33] Composites were prepared by homogenizing equally weighted aliquots from each fish by station, species, and size class. A fish sample was analyzed as an individual for common carp, as this species was difficult to catch during sample cruises. Typically, samples were then shipped to the lab and frozen at −20 °C until analysis within 90 days. A subset of samples across all monitoring years were archived at −20 °C before analysis in 2022. These archived samples are typically considered together with samples collected and analyzed in 2009, 2014, and 2019 unless otherwise indicated. The Supporting Information (SI: Tables S1 and S2) presents additional details on individual fish species, sampling, and composites.

Analytical Methods. From 2009 onward, SGS AXYS Analytical Services Ltd. (Victoria, BC, Canada) conducted analyses of all fish tissue samples (2.0 g of homogenized material). The methods used are accredited by NELAP and IS0 17025, utilizing liquid chromatography coupled with mass spectrometry (LC−MS/MS) instruments that apply isotope dilution/recovery correction quantification.

Analysis in 2009 and 2014 consisted of nine PFCAs (PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA), three PFSAs (PFBS, PFHxS, PFOS), and perfluorooctane sulfonamide (PFOSA), a PFOS precursor (SI: Table S3). Continued improvements in analytical methods and available analytical standards and labeled surrogates have expanded the breadth of PFAS detection with the addition of 20 analytes in 2019 and another 7 in 2022 (40 total) including two additional PFCAs (PFTrDA, PFTeDA), five additional PFSAs (PFPeS, PFHpS, PFNS, PFDS, PFDoS), and 20 PFAS precursors [three fluorotelomer sulfonates (FTSAs: 4:2, 6:2, 8:2), three fluorotelomer carboxylates (FTCAs: 3:3, 5:3, 7:3), two other perfluorooctane sulfonamides (FOSAs: N-MeFOSA, N-EtFOSA), two perfluorooctane sulfonamide acetic acids (FOSAAs: N-MeFOSAA, N-EtFOSAA), two perfluorooctane sulfonamide ethanols (FOSEs: N-MeFOSE, N-EtFOSE), five per- and polyfluoroether carboxylic acids (PFECAs: HFPO-DA, ADONA, NFDHA, PFMBA, PFMPA), two chlorinated polyfluoroalkyl ether sulfonic acids (Cl-PFESAs: 9Cl-PF3ONS; 11-Cl-PF3OUdS), and one perfluoroalkyl ether sulfonate (PFESA: PFEESA)] (SI: Tables S3 and S4).

Overall protocols, instrumentation, and quantification methodology have changed only slightly for PFCAs and PFSAs since 2009. Sample analysis in 2009, 2014, and 2019 was conducted using internal methods developed by SGS AXYS similar to the protocols in EPA Method 1633, which was used for sample analysis in 2022. Through these methods, samples were spiked prior to extraction, with isotope-labeled surrogate standards, extracts were cleaned and concentrated via weak ion exchange (WAX) solid phase extraction (SPE) and subsequently analyzed via LC−MS/MS. Several multiple reaction monitorings (MRMs) have been monitored for each compound where available. The number of available labeled surrogates has expanded significantly since 2009, culminating in those surrogates, MRM transitions, and MRM transition ratios published in EPA Method 1633.

Analytes were quantified using the isotope dilution-internal standard method with multipoint calibration curves. All results are reported to the lowest calibration point, noted as the reporting limit (RL), prorated for mass. A statistically calculated method detection limit (MDL) below that calibration point is applied for samples analyzed in 2022. RLs have consistently decreased over time with an order of magnitude difference between RLs for analytes measured in 2009 (range: 2.4−5 ng/g ww) to 2022 (range: 0.1−0.44 ng/g ww; SI: Table S5).

Standard quality assurance/quality control (QA/QC) procedures were used, including field duplicates, lab duplicates, lab control spikes, method blanks, matrix spikes (MS), and matrix spike duplicates (MSD) with relevant information presented in the SI (Tables S6−S8). Particularly, method blanks across batches generally did not detect any PFAS in 2009 and 2014. Five analytes (PFUnDA, PFTrDA, PFOS, 6:2 FTS, and PFOSA) were detected at low levels in various method blanks from 2019 and 2022 analyses (SI: Table S8), with sample concentrations in the same batch marked nondetect if the concentration was less than three times the blank concentration. Assessment of method trueness through analysis of standard reference material (NIST SRM 1947) showed relative percent differences (RPDs) of ≤20% compared to the reference concentrations of PFOS (SI: Table S9).

Several archived samples analyzed in 2022 were duplicates of samples previously collected and examined in 2009, 2014, or 2019 (SI: Tables S10−S12). PFOS concentrations notably increased for the two samples from 2009 reanalyzed in 2022 using EPA Method 1633, which could be largely attributed to the lower RLs achieved. Three long-chain PFAS (PFNA, PFDA, and PFDoDA) were included in both methods, but detected at low concentrations only in 2022, for the same reason as PFOS. These differences could also be attributed to the potential degradation of PFAS precursors to terminal products over time in archived samples and the potential error associated with measuring a sample with varying methods and equipment. An additional five PFAS, included only in the 2022 method, were also found in archived samples from 2009. The eight samples from 2014 showed generally similar levels of PFOS upon reanalysis, with RPDs mostly below ≤15%. PFOA, PFNA, and PFHxS were a part of both analyses though found only in the 2022 methods at low levels likely due to the lower RLs. Six PFAS only included in the 2022 analysis were detected in these samples, with precursor 7:3 FTCA detected at particularly high levels. The two 2019 samples reanalyzed in 2022 had similar PFOS concentrations (RPDs ≤ 15%) with low detections of three long-chain PFAS (two of which are newly included). In the following analysis, only the reanalyzed 2022 archived sample duplicates are included when data sets are combined across different analysis years.

Average concentrations of PFOS presented per species were calculated by using half the RL for nondetects of samples analyzed using the 2019 and 2022 methods; otherwise,

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

Table 1. Reporting Limits, Detection Frequency, and Range of Detected Concentrations of PFOS (ng/g ww) in Fish Samples from 2009, 2014, and 2019, Including the Percentage of Samples above MA Consumption Advisory Thresholds

	2009[a] (n = 21)	2009/2022[a] (n = 7)	2014[b] (n = 36)	2019 (n = 33)	total (n = 95)[c]
reporting limits (ng/g)	4.8−5	0.1	0.1−1	0.1−0.2	
detection frequency (%)	19	100	94	91	
range of detected PFOS concentrations (ng/g ww)	5.6−18	2.8−8.5	0.9−17	0.59−10	
samples above the MA one meal per day consumption advisory threshold for the general population: ≤0.50 ng/g	19%	100%	94%	91%	83%
samples above the MA one meal per week consumption advisory threshold for the general population: ≤3.52 ng/g	19%	57%	36%	27%	32%

[a]Due to differences in method sensitivity, samples collected and analyzed in 2009 were separated from archive samples analyzed in 2022 (labeled 2009/2022 above). For other monitoring years, all sample data was combined. [b]White croaker (n = 7) from 2014 were processed whole (skin on) without head or guts, which is different from other years where they were processed as fillet with skin off. [c]Two samples analyzed in 2009 were excluded as the reanalyzed samples from 2022 were used instead.

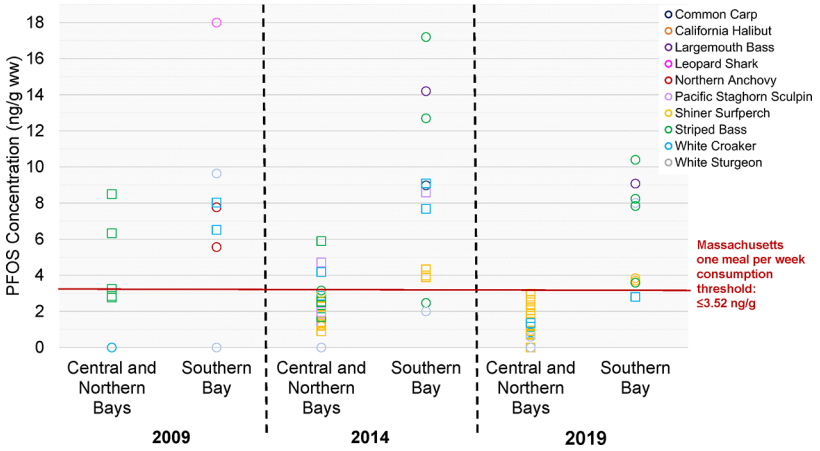


Figure 2. Comparison of concentrations of PFOS from 2009 to 2019 in fish caught in the southern San Francisco Bay and the central and north San Francisco Bay sites. Circles indicate fish analyzed in the same monitoring year, and squares note those archived and analyzed in 2022.

nondetects were considered zero due to the relatively high RLs of the other methods.[34] For Sum PFAS, analytes detected in greater than 70% of samples had nondetects replaced by half the RL for samples analyzed in 2019 and 2022.[34]

RESULTS AND DISCUSSION

PFOS: Occurrence, Spatial Trends, and Risk. PFOS was the predominant PFAS found in San Francisco Bay fish and was detected in ≥91% of the samples across analytical methods (Table 1). An exception were those San Francisco Bay fish samples collected and analyzed in 2009 (detection frequency = 19%; n = 21), likely due to the higher RLs associated with the early method used at that time (median RL = 4.9 ng/g ww). Analysis of archived 2009 samples (n = 7) conducted in 2022 using EPA Method 1633 (median RL = 0.1 ng/g ww) found all contained PFOS, with three samples having concentrations below the 2009 RL. As the 2009 method was less sensitive than EPA Method 1633, concentrations of PFOS and related summary statistics were considered separately for fish sampled and analyzed in 2009 versus archived samples analyzed in 2022. This distinction was not necessary for samples collected in 2014 and 2019, as method sensitivity for PFOS was similar whether analyses were conducted during the collection year or later in 2022. The complexity of the analytical methods, varying processing methods, and limited sample sizes across monitoring years meant we were not able to statistically assess differences in PFAS concentrations over time.

Thresholds and consumption guideline values for PFOS in fish tissue vary by location. In the EU, the Environmental Quality Standard (EQS) in the Water Framework Directive for PFOS in biota is set for protecting human health at 9.1 ng/g.[35] This value was derived using a tolerable daily intake (TDI) developed by the European Food Safety Authority (EFSA) in 2008.[36] In 2020, EFSA published a revised, significantly lower TDI for the sum of PFOS, PFOA, PFNA, and PFHxS based on the immunological effects of these compounds and the association between PFAS exposure and reduced vaccine response, but the EQS has not been updated.[23] Langberg et al. calculated that a new EQS for biota protective of human health would be 0.04 ng/g using this updated TDI.[35]

In the U.S., state fish consumption guidelines vary dramatically due to differences in reference dose values used to determine the guidelines, which have generally diminished over time as more toxicity data have indicated adverse effects at lower exposure levels.[24] Using the fish tissue thresholds developed by the state of Massachusetts,[37] the most stringent in the U.S., 83% of all San Francisco Bay fish samples were above the one meal per day (eight-ounce serving) consumption advisory threshold for the general population (≤0.50 ng/g), and 32% of all samples were above the one meal per week consumption threshold (≤3.52 ng/g, Table 1). Of the sampled species, 5 of 7 in 2009 and 2014 had a PFOS exceedance of the one meal per day threshold and 4 of 5 exceeded in 2019 (Figures S1−S3). Four species in 2009 (range of species averages: 4.4−8 ng/g ww), five in 2014

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

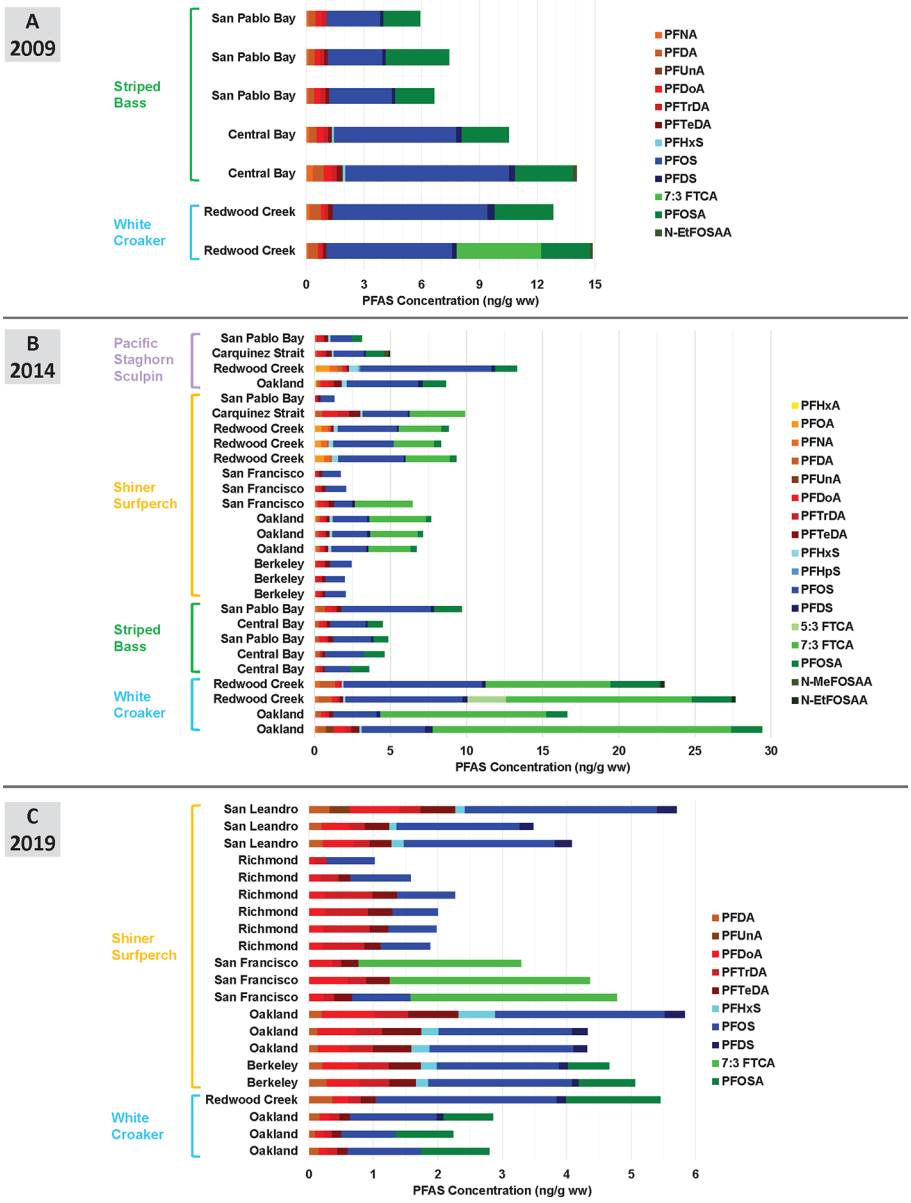


Figure 3. Concentrations of all PFAS detected in archived fish samples in 2009 (A), 2014 (B), and 2019 (C) using EPA Method 1633 in 2022.

(range of species averages: 4.2−14 ng/g ww), and two in 2019 (range of species averages: 7.5−9.1 ng/g ww) had average PFOS concentrations above the Massachusetts one meal per week advisory threshold. Due to the wide variance in RLs between methods across monitoring years, nondetects were considered zero in the calculation of these species averages. Though sample processing methods varied across fish, with some whole fish samples analyzed, these thresholds are the best available comparison and highlight a need for thresholds inclusive of more than fish fillets.

Thresholds for ecotoxicity of PFOS, intended to protect the fish themselves or mammalian or avian species that consume fish, have also been developed. These values are generally higher than those intended to protect human health. For example, the Canadian Federal Environmental Quality Guidelines for PFOS are 9400, 4.6, and 8.2 ng/g ww for protection of fish, mammalian wildlife, and avian wildlife, respectively.[38] None of the samples from San Francisco Bay exceeded this threshold for fish tissue, though 26% and 13% of samples were above the mammalian wildlife and avian wildlife thresholds, respectively.

Fish caught in the southern San Francisco Bay (Coyote Creek, Redwood Creek, and Artesian Slough sites; Figure 1) generally had the highest PFAS concentrations and greater PFAS detections. This pattern remained consistent in all monitoring years and among most species examined (Figure 2). PFOS was particularly elevated in the southern San Francisco Bay, with 90% of sampled fish (detected range: 2.0−18 ng/g ww) exceeding the Massachusetts one meal per day consumption threshold compared to 77% (detected range: 0.59−8.5 ng/g ww) in the central and northern San Francisco Bay (Berkeley, Carquinez Strait, Central Bay, Oakland, Richmond, San Leandro Bay, San Pablo Bay, San Francisco, and Suisun Bay; Figure 1). Similarly, 81% of southern San Francisco Bay fish exceeded the one meal per week consumption threshold for PFOS, compared to 8%, in the central and northern San Francisco Bay. Concentrations of PFOS in the southern San Francisco Bay fish remained

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

consistently high over time, which contrasts with significant declines observed in other biota in the region, including bird eggs and harbor seals.[30] Even with the differences in total PFAS analyzed in each monitoring year due to evolving methods, southern San Francisco Bay fish also showed the greatest Sum PFAS (Figure 2).

Higher levels of PFAS in the southern San Francisco Bay are partially attributed to limited hydraulic flushing relative to other San Francisco Bay regions, increasing the influence of PFAS discharges.[39] Potential sources and pathways of PFAS to receiving waters are prevalent across the San Francisco Bay Area due to its highly urbanized landscape, with the southern San Francisco Bay region highly influenced by municipal wastewater effluents and urban stormwater compared to other regions.[40,41] Another key difference in the southern San Francisco Bay is that it only receives a small portion (~10%) of the watershed flows coming into the Bay compared to other regions, and it experiences limited tidal flushing from the Pacific Ocean.[39] Measurements of PFAS in other biota, surface water, and sediment have also shown higher levels of PFAS in the southern San Francisco Bay.[42,43]

Similar studies of marine and freshwater fish tissue globally have also generally shown PFOS as the predominant PFAS with reported concentrations ranging from a few ng/g to over 200 ng/g.[13,35,46−48] Overall, levels of PFOS in San Francisco Bay fish have been generally within range or lower than those in other studies across the U.S. where anticipated sources and regulation of PFAS are comparable. In marine fish from Charleston Harbor, South Carolina (2014), PFOS concentrations ranged from 0.40 to 30 ng/g ww, similar to the range observed in San Francisco Bay fish during the same sample period (range: <MDL−17 ng/g ww).[25] Marine fish from Rhode Island from 2006 to 2014 were noticeably different from most studies as PFOS were not the dominant PFAS detected and generally found at low concentrations (range: <MDL−7.5 ng/g ww).[14] The concentrations of PFOS in Tampa Bay, Florida (2021−2022) had a greater range, from 0.92 to 30 ng/g ww, than those in 2019 San Francisco Bay fish (range: 0.59−10 ng/g ww).[28] Freshwater fish generally showed PFOS concentrations that ranged higher relative to similar sampled years in the San Francisco Bay, including the U.S. Great Lakes in 2015 (range: <MDL−64 ng/g ww), U.S. rivers and streams from 2013 to 2015 (range: <MDL−283 ng/g ww), New Jersey in 2015−2016 (range: <MDL−208 ng/g), North Carolina in 2020 (range: 3.6−130 ng/g ww), and the New Hampshire Merrimack rivershed in 2019 (range: 0.21−52 ng/g ww).[18,24,44,45]

Occurrence of PFAS Precursors and Long-Chain PFCAs. Along with PFOS, an additional 19 PFAS were observed in San Francisco Bay fish (Tables S5−S7). An examination of the PFAS profiles for each archived sample, those analyzed using the expanded method in 2022, across all collection years has identified two distinct groups of PFAS outside of PFOS that primarily contribute to overall PFAS levels. The group most significantly adding to the PFAS concentrations in fish is PFAS precursors (in green in Figure 3), primarily PFOSA and 7:3 FTCA. A PFOS precursor, PFOSA was frequently detected in 2009 (detection frequency: 100%; detected range: 1.9−3.0 ng/g ww) and 2014 (detected frequency: 70%; detected range: 0.34−3.2 ng/g ww). However, in 2019, only 29% of samples (detected range: 0.64−1.5 ng/g ww) contained PFOSA, which could be an effect of the detection of PFOSA in a lab blank (resulting in censoring of low-level detections in 8 of 27 samples). Two additional PFOS precursors, N-MeFOSAA and N-EtFOSAA, were found in a few samples in each collection year at low levels (≤0.30 ng/g ww). Higher trophic level species, striped bass and white croaker (trophic level 4), generally had more frequent detections and higher concentrations of PFOS precursors relative to the lower trophic level species, shiner surfperch (trophic level 3).

PFOS precursors, especially PFOSA, have been widely found in fish and other aquatic life with marine mammals showing the highest levels, one to two orders of magnitude greater than in fish.[8,13,18,24,25,45,49−53] Several studies have noted the limited capacity of marine mammals to biotransform PFOSA to PFOS, with comparisons of ratios of PFOS/PFOSA indicating varying rates of transformation across species and size.[52,53] A similar analysis of fish fillet samples in Germany found varying biotransformation capacity among fish species, with higher ratios of PFOS/PFOSA in larger species.[13] This study supports the variability seen among fish species when comparing shiner surperch (trophic level 3; ratio range: 6.1−10.5) to striped bass (trophic level 4; ratio range: 0.9−6.1). However, ratios were higher in shiner surfperch, the smaller species, which is in contrast to what was observed in German fish. The difference of biotransformation of PFOS precursors across fish species in the San Francisco Bay could be one of the potential explanations for the varying concentrations seen across species.

Notably, 7:3 FTCA, a PFCA precursor that was first incorporated into analyses in 2022, was found to be a dominant PFAS detected in many samples (Figure 3.) In 2009 and 2019, 14% of archived samples had relatively high levels of 7:3 FTCA (range: 2.5−3.8 ng/g ww), and even greater amounts were observed in archived samples collected in 2014 (detection frequency: 44%; range: 2.6−20 ng/g ww). Only two species, white croaker and shiner surfperch, had detectable concentrations of 7:3 FTCA, with no other clear patterns or established sources for the detections. There are no known consumer or industrial applications of 7:3 FTCA, with detections linked to the compound being a transformation product from other larger PFAS such as 8:2 fluorotelomer alcohol and fluorotelomer acrylates in fish.[54,55]

This is the first published detection of 7:3 FTCA in marine fish. The concentrations of 7:3 FTCA were generally higher than the few available studies of freshwater fish around Michigan, eastern Canada, and Hong Kong (range: 0.07−4.1 ng/g ww).[12,56,57] 7:3 FTCA has also been reported in both seabirds and marine mammals globally, highlighting its remarkable bioaccumulation potential in marine organisms and/or the significant presence of unknown precursors degrading to this compound.[8,49−52,58] One study on the U.S. Atlantic Coast found 7:3 FTCA to be the dominant PFAS in livers of marine mammals, with concentrations in seals (maximum: 192 ng/g ww) and polar bears (maximum: 1131 ng/g ww) one to 2 orders of magnitude greater than those in San Francisco Bay fish.[8] To the best of our knowledge, there are currently no thresholds or guidelines for 7:3 FTCA in fish tissues. In general, both ecotoxicity and human health thresholds for PFAS precursors have not been developed (often due to a lack of toxicity data), representing a key data gap for water quality management.

The next most widely observed group of PFAS was long-chain PFCAs, with a total of seven found from 2009 to 2019 (in red in Figure 3). PFCAs with the longest chains, PFDoDA,

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

PFTeDA, and PFTrDA (the latter two first analyzed in 2022), were found in greater than 86% of all archived samples, though at generally low concentrations (≤0.79 ng/g ww). PFNA and PFDA were also found in multiple samples. However, their detection frequency (and overall contribution to PFAS levels) has decreased over time, from 100% in 2009 for both to 0% and 57% in 2019 for PFNA and PFDA, respectively, and concentrations were also generally low (≤0.79 ng/g ww). Additionally, PFOA and PFUnDA were detected sporadically at low levels from 2009 to 2019. When combined, long-chain PFCAs comprise a large portion of the overall PFAS concentrations in San Francisco Bay fish.

Detections of many of these PFCAs and other detected PFAS have been reported in studies of freshwater and marine fish at low concentrations in the U.S. and internationally.[13,18,24−28,44−48] Advisory thresholds from Massachusetts for both sensitive and general populations for PFOA (maximum: 0.90 ng/g), PFNA (maximum: 0.50 ng/g), PFBS (all <RL), and PFHxS (maximum: 0.57 ng/g) were not exceeded in any samples. Newer PFAS, such as HFPO-DA (GenX) and ADONA, were not detected in any fish.

Assessing risk based on exposure to total PFAS—as opposed to evaluating each compound individually—is increasingly recognized as necessary. The high persistence, accumulation potential, and hazards (known and potential) of PFAS studied to date warrant treating all PFAS as a single class;[59] however, few thresholds have been developed for Sum PFAS, and those that have been developed are generally for only a handful of long-chain compounds. In 2022, the European Commission set a general maximum level for the sum of PFOS, PFOA, PFNA, and PFHxS in fish muscle of 2 ng/g ww.[60] All of the archived samples from 2009 and 67% of the archived samples from 2014 exceeded this limit. Of all of the samples in 2019 (including both those analyzed in 2019 and 2022), 48% of samples were above this level. All of the archived samples from 2009 and 2014 along with all of the 2019 samples from the South Bay exceeded this threshold. Langberg et al. suggest that a quality standard using EFSA's updated TWI would be 0.04 ng/g for the same set of compounds. Every San Francisco Bay fish sample had a sum of PFOS, PFOA, PFNA, and PFHxS higher than this suggested threshold.[35]

An EQS of 0.077 ng/g ww has been proposed for the sum of 24 PFAS expressed as PFOA toxicity equivalents, based on the EFSA TDI and relative potency factors (RPFs) derived from hepatotoxic potency in rats.[60] However, this value has received critique, as the RPFs have not been validated.[61] The Sum PFAS in archived samples were variable throughout monitoring years and across sampled species, with concentrations ranging from 5.9 to 15 ng/g ww in 2009 (n = 7; 12 PFAS detected; Table S10), 1.3 to 29 ng/g ww in 2014 (n = 27; 17 PFAS detected; Table S11), and 0.15 to 15 ng/g ww in 2019 (n = 33; 15 PFAS detected, Table S12).

Observations in Individual Species. Three species of fish (shiner surfperch, striped bass, and white croaker) were sampled more frequently in each monitoring year to provide further insight into PFAS levels and patterns from 2009 to 2019. These species were selected for monitoring to encompass a range of feeding habits, home ranges, and lifespans (SI: Table S2). Distinct differences emerged from a species-specific review of the data, as detailed below.

Shiner Surfperch. Found along the shoreline and piers of San Francisco Bay, shiner surfperch are small fish (up to 18 cm) that feed on benthic organisms and are commonly caught by recreational and subsistence fishers. Due to their small home ranges (<10 km), tissue concentrations are more clearly tied to local PFAS contamination at a sampled site.[31]

Of the three shiner surfperch samples analyzed in 2009, none had detectable PFAS, likely due to the high RLs (2.5−5 ng/g). PFOS was the predominant PFAS found in 2014 (n = 14; detection frequency: 100%) and 2019 (n = 20; detection frequency: 88%) with comparable averages of 2.2 and 1.8 ng/g ww, respectively (Figure S4). At sites sampled in both years, PFOS levels were also similar. PFOS concentrations in 86% of samples were above the one meal per day Massachusetts threshold (≤0.5 ng/g) and 16% of samples were above the one meal per week threshold (≤3.5 ng/g). All of the samples above the one meal per week threshold were from southern San Francisco Bay sites, further highlighting the elevated concentrations in this region. The shiner surfperch diet of benthic organisms could be a contributing factor, with PFOS predominantly present in sediment across San Francisco Bay, especially in the southern regions.[43] A previous 2012 study of prey fish in San Francisco Bay found distinctly higher levels of PFOS (range: 1.4−29 ng/g ww) in shiner surfperch, which may be related to the selection of different sites that were chosen to reflect proximity to harbor seal haulouts rather than human fishing activity.[30]

An additional nine PFAS were detected in samples of shiner surfperch in 2014 and 2019, with PFDoDA, PFTrDA, and PFTeDA being the only analytes found in ≥70% of analyzed samples. Average concentrations of PFDoA, PFTrDA, and PFTeDA ranged from 0.2 to 0.4 ng/g ww. Unlike PFOS, there were no consistent spatial trends for concentrations of these other PFAS in shiner surfperch. In fact, a sample from Carquinez Strait, in the northern part of San Francisco Bay, had the highest levels of all three of these PFCAs in 2014. Local inputs including urban stormwater, wastewater effluent, and nearby oil refineries' wastewater in the northern San Francisco Bay could be contributing to these elevated levels.

When 7:3 FTCA was detected in archived samples (2014 detection frequency: 53%; 2019 detection frequency: 17%), it was at relatively high levels (≥2.5 ng/g ww), contributing greater than a third of the total PFAS signal in shiner surfperch samples. Only at the San Francisco station in the Central Bay was 7:3 FTCA detected in both 2014 and 2019, although there are no known sources to attribute to these findings.

Striped Bass. Striped bass are another species commonly caught and consumed by recreational and subsistence fishers in San Francisco Bay. During the summer and fall, they are found throughout the Bay with some migrating to the ocean or to freshwater spawning grounds in the Delta region during the winter and spring. Typically, striped bass caught by anglers are in the 50−80 cm range. Striped bass in the Bay feed on invertebrates and smaller fish, including shiner surfperch.[31]

All archived striped bass samples from 2009 (n = 5) analyzed using EPA Method 1633, as well as all data from 2014 (n = 9) and 2019 (n = 4), had PFOS detections, with average concentrations of 4.7 ng/g ww in 2009, 5.6 ng/g ww in 2014, and 7.5 ng/g ww in 2019 (Figure S5). Higher concentrations in later years may be due to the inclusion of southern Bay samples (all of the 2019 striped bass samples were from the southern Bay), which have been consistently higher compared with other parts of the Bay. PFOS concentrations in 100% of samples were above the one meal per day MA threshold (≤0.5 ng/g) and 50% of samples were above the one meal per week threshold (≤3.5 ng/g). The higher frequency of exceedances,

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

especially in the one meal per week threshold, and higher overall average concentrations when compared to shiner surfperch could be explained by the higher collection frequency of striped bass near the San José-Santa Clara wastewater outfall, which is a known PFAS pathway. Another possible explanation is the potential biomagnification of PFOS from shiner surfperch (trophic level 3) to striped bass (trophic level 4). Biomagnification of PFOS has been observed in other environments for fish and other aquatic organisms, including a pelagic coastal North Atlantic food web with striped bass.[8−14] Further studies are needed to understand biomagnification in the San Francisco Bay and the overall importance of varying environmental and biological factors in the presence of PFAS.

PFOSA and PFTeDA were also frequently detected (≥75%) in striped bass across all sample years. After PFOS, the next highest concentrations were generally found for PFOSA, with average concentrations remaining relatively consistent over time from 1.8 ng/g ww in 2009, 1.2 ng/g in 2014, and 1.5 ng/g in 2019. In 2009 and 2014, PFTrDA was found in all samples at low levels (around 0.2 ng/g), though in 2019 only 50% of samples had a detection.

PFNA, PFDA, PFUnDA, PFHxS, PFDS, and Et-FOSAA, a PFSA precursor, were detected more sporadically across sample years, although notably all except PFHxS were found in 2019 samples in ≥75% of samples. This helps explain the elevated Sum PFAS in 2019 compared with other sample years (Table S12) and further underscores the elevated levels of contamination in the southern San Francisco Bay, where all of the 2019 striped bass samples were collected. Overall, PFAS, especially PFOS, appear to be persisting at relatively high levels in striped bass.

White Croaker. White croaker is another species commonly fished in the San Francisco Bay and analyzed in all sample years. White croaker have a large and variable home range with migration toward deeper water and the ocean during the winter. White croaker caught by anglers typically range from 200 to 300 cm. White croaker feed on benthic organisms.[31]

All samples of white croaker analyzed using EPA Method 1633 from 2009 (n = 2), 2014 (n = 4), and 2019 (n = 5) were found to contain PFOS. The average concentrations of PFOS were 7.3, 6, and 1.4 ng/g ww in 2009, 2014, and 2019, respectively (Figure S6). It is important to note that samples from 2014 were processed whole (without head or guts; with skin on), compared to skinless fillet samples in other years, which could influence PFAS levels. In addition, the small sample sizes limit robust analysis of trends in white croaker. Still, it is important to note that PFOS concentrations in all white croaker samples were above the one meal per day MA threshold (≤0.5 ng/g) and 46% of samples were above the one meal per week threshold (≤3.5 ng/g).

Similar to striped bass, PFOSA and PFTeDA were commonly detected in white croaker across all sample years (detection frequency: ≥75%), with the former also being detected in the second greatest concentrations (after PFOS) across sample years (range: 0.7−3.2 ng/g ww). An additional 12 PFAS were found, including several precursor compounds: 5:3 FTCA, 7:3 FTCA, Me-FOSAA, and Et-FOSAA. The detections of precursors mostly occurred in 2014 samples, which is likely due to the use of whole fish samples compared to fillet samples in other monitoring years. 7:3 FTCA was detected in several samples at high levels (range: 4.3−20 ng/g), approximately double the concentrations in shiner surfperch caught at the same stations.

The Sum PFAS in white croaker in 2009 and 2014 were comparatively high relative to other species, especially in 2014 where two fish had the highest levels recorded in any San Francisco Bay fish, near 30 ng/g ww (Table S11). Sum PFAS values were much lower in 2019.

Implications. This work highlights the widespread contamination of PFAS in fish from an urban estuary across a decade of monitoring with important public health implications. San Francisco Bay fish are likely a key source of exposure to PFOS and other PFAS for those who consume them, with more significant impacts to subsistence fishers and others with high consumption rates. PFOS contamination persists in San Francisco Bay fish above stringent consumption advisory thresholds, despite management actions to reduce its production and use. The concern is greatest in the southern San Francisco Bay, where concentrations have been consistently higher than in other regions of the Bay.

Our findings indicate a need for sustained monitoring to protect public health and provide more rigorous data to evaluate temporal trends. Diverse patterns of detection dependent on species and sites indicate further study of similar species across San Francisco Bay's subregions is essential to effectively monitor trends and guide management actions. These efforts should be paired with the monitoring of apex predator species that consume fish as well as water and sediment to better understand the ecological impacts of PFAS. While PFOS remains the compound of greatest concern in San Francisco Bay fish, emerging detection of 7:3 FTCA and other precursors indicates a continued need to analyze fish samples using methods that assess a broad array of PFAS, especially the inclusion of suspect screening and nontargeted analyses. This suggests the need for exposure and risk evaluations for the many PFAS consistently found in San Francisco Bay fish samples as well as monitoring of potential PFAS sources and pathways to understand and minimize their emissions into San Francisco Bay. Concerns highlighted here likely extend to fish caught in other urban water bodies around the world.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestwater.4c00999.

Detailed figures of average concentrations of PFOS highlighting all species sampled in each monitoring year and temporal trends of the three species sampled every monitoring period; a diagram of the different methods used (PDF)

Further fish sampling and species biological information; supporting descriptions of methods analyses including specific analytes, reporting limits, and recovery ranges for surrogate and spiked samples; blank and CRM concentrations; and tables of fish concentrations (XLSX)

■ AUTHOR INFORMATION

Corresponding Author

Rebecca Sutton − *San Francisco Estuary Institute, Richmond, California 94804, United States;* orcid.org/0000-0002-8388-8222; Email: rebeccas@sfei.org

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913

Authors

Miguel A. Méndez − *San Francisco Estuary Institute, Richmond, California 94804, United States;* orcid.org/0009-0006-1915-3340

Jay Davis − *San Francisco Estuary Institute, Richmond, California 94804, United States*

Ezra L. Miller − *San Francisco Estuary Institute, Richmond, California 94804, United States;* orcid.org/0000-0002-9711-2863

Richard Grace − *SGS AXYS Analytical Services, Ltd., Sidney, BC V8L 5X2, Canada*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsestwater.4c00999

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Funding for this project was provided by the Regional Monitoring Program for Water Quality in San Francisco Bay and the San Francisco Bay Regional Water Quality Control Board. The authors thank SFEI staff including Martin Trinh for archived sample retrieval, Adam Wong and Donald Yee for data services and quality assurance review, and Regan Murray for GIS services. Additional thanks to Paul Salop of Applied Marine Sciences, Inc. for his many years of assistance with the RMP fish archives. The authors also thank Marco Sigala, Autumn Bonnema, and the entire team at Marine Pollution Studies Lab at Moss Landing Marine Laboratories for their help in sampling and processing fish throughout the years. Thank you to Dr. Derek Muir (Environment and Climate Change Canada), Dr. William Arnold (University of Minnesota), Dr. Simona Balan (California Department of Toxic Substances), Dr. Dan Villenueve (USEPA), Dr. Wesley Smith (California Office of Environmental Health and Hazard Assessment), and Drs. Richard Looker and Thomas Mumley (San Francisco Bay Regional Water Quality Control Board) for their review of this work. The authors acknowledge the Indigenous Peoples whose unceded lands surround San Francisco Bay, including the Him-R^n Ohlone Jalquin, Saclan Tribe, the Villages of Lisjan, the Karkin, Muwekma, Ramaytush, Tamien, and Yokuts Ohlone, the Coast and Bay Miwok, Patwin, and the Amah Mutsun Tribal Band.

■ REFERENCES

(1) Dewapriya, P.; Chadwick, L.; Gorji, S. G.; Schulze, B.; Valsecchi, S.; Samanipour, S.; Thomas, K. V.; Kaserzon, S. L. Per- and Polyfluoroalkyl Substances (PFAS) in Consumer Products: Current Knowledge and Research Gaps. *J. Hazard. Mater. Lett.* **2023**, *4*, No. 100086.

(2) Glüge, J.; Scheringer, M.; Cousins, I. T.; DeWitt, J. C.; Goldenman, G.; Herzke, D.; Lohmann, R.; Ng, C. A.; Trier, X.; Wang, Z. An Overview of the Uses of Per- and Polyfluoroalkyl Substances (PFAS). *Environ. Sci.: Processes Impacts* **2020**, *3*, 1243−1259, DOI: 10.1039/D0EM00291G.

(3) USEPA. Our Current Understanding of the Human Health and Environmental Risks of PFAS. United States Environmental Protection Agency, 2022. https://www.epa.gov/pfas/our-current-understanding-human-health-and-environmental-risks-pfas.

(4) Cousins, I. T.; Johansson, J. H.; Salter, M. E.; Sha, B.; Scheringer, M. Outside the Safe Operating Space of a New Planetary Boundary for Per- and Polyfluoroalkyl Substances (PFAS). *Environ. Sci. Technol.* **2022**, *56* (16), 11172−11179.

(5) Evich, M. G.; Davis, M. J. B.; McCord, J. P.; Acrey, B.; Awkerman, J. A.; Knappe, D. R. U.; Lindstrom, A. B.; Speth, T. F.; Tebes-Stevens, C.; Strynar, M. J.; Wang, Z.; Weber, E. J.; Henderson, W. M.; Washington, J. W. Per- and Polyfluoroalkyl Substances in the Environment. *Science* **2022**, *375* (6580), No. eabg9065.

(6) Khan, B.; Burgess, R. M.; Cantwell, M. G. Occurrence and Bioaccumulation Patterns of Per- and Polyfluoroalkyl Substances (PFAS) in the Marine Environment. *ACS EST Water* **2023**, *3*, 1243−1259, DOI: 10.1021/acsestwater.2c00296.

(7) Agency for Toxic; Substances and Disease Registry. *Toxicological Profile for Perfluoroalkyls*; US Department of Health and Human Services: Atlanta, GA, 2021.

(8) Spaan, K. M.; van Noordenburg, C.; Plassmann, M. M.; Schultes, L.; Shaw, S.; Berger, M.; Heide-Jørgensen, M. P.; Rosing-Asvid, A.; Granquist, S. M.; Dietz, R.; Sonne, C.; Rigét, F.; Roos, A.; Benskin, J. P. Fluorine Mass Balance and Suspect Screening in Marine Mammals from the Northern Hemisphere. *Environ. Sci. Technol.* **2020**, *54* (7), 4046−4058.

(9) Miranda, D. A.; Zachritz, A. M.; Whitehead, H. D.; Cressman, S. R.; Peaslee, G. F.; Lamberti, G. A. Occurrence and Biomagnification of Perfluoroalkyl Substances (PFAS) in Lake Michigan Fishes. *Sci. Total Environ.* **2023**, *895*, No. 164903.

(10) Miranda, D. de A.; Peaslee, G. F.; Zachritz, A. M.; Lamberti, G. A. A Worldwide Evaluation of Trophic Magnification of Per- and Polyfluoroalkyl Substances in Aquatic Ecosystems. *Integr. Environ. Assess. Manage.* **2022**, *18* (6), 1500−1512.

(11) Munoz, G.; Mercier, L.; Duy, S. V.; Liu, J.; Sauvé, S.; Houde, M. Bioaccumulation and Trophic Magnification of Emerging and Legacy Per- and Polyfluoroalkyl Substances (PFAS) in a St. Lawrence River Food Web. *Environ. Pollut.* **2022**, *309*, No. 119739.

(12) George, S. E.; Baker, T. R.; Baker, B. B. Nonlethal Detection of PFAS Bioaccumulation and Biomagnification within Fishes in an Urban- and Wastewater-Dominant Great Lakes Watershed. *Environ. Pollut.* **2023**, *321*, No. 121123.

(13) Babut, M.; Labadie, P.; Simonnet-Laprade, C.; Munoz, G.; Roger, M.-C.; Ferrari, B. J. D.; Budzinski, H.; Sivade, E. Per- and Poly-Fluoroalkyl Compounds in Freshwater Fish from the Rhône River: Influence of Fish Size, Diet, Prey Contamination and Biotransformation. *Sci. Total Environ.* **2017**, *605*−*606*, 38−47.

(14) Hedgespeth, M. L.; Taylor, D. L.; Balint, S.; Schwartz, M.; Cantwell, M. G. Ecological Characteristics Impact PFAS Concentrations in a U.S. North Atlantic Food Web. *Sci. Total Environ.* **2023**, *880*, No. 163302.

(15) Brendel, S.; Fetter, É.; Staude, C.; Vierke, L.; Biegel-Engler, A. Short-Chain Perfluoroalkyl Acids: Environmental Concerns and a Regulatory Strategy under REACH. *Environ. Sci. Eur.* **2018**, *30* (1), 1−11, DOI: 10.1186/s12302-018-0134-4.

(16) KEMI Swedish Chemicals Agency. *Occurrence and Use of Highly Fluorinated Substances and Alternatives*; Swedish Chemicals Agency: Stockholm, 2015.

(17) Wang, Z.; Buser, A. M.; Cousins, I. T.; Demattio, S.; Drost, W.; Johansson, O.; Ohno, K.; Patlewicz, G.; Richard, A. M.; Walker, G. W.; White, G. S.; Leinala, E. A New OECD Definition for Per- and Polyfluoroalkyl Substances. *Environ. Sci. Technol.* **2021**, *55* (23), 15575−15578.

(18) Pickard, H. M.; Ruyle, B. J.; Thackray, C. P.; Chovancova, A.; Dassuncao, C.; Becanova, J.; Vojta, S.; Lohmann, R.; Sunderland, E. M. PFAS and Precursor Bioaccumulation in Freshwater Recreational Fish: Implications for Fish Advisories. *Environ. Sci. Technol.* **2022**, *56*, No. acs.est.2c03734.

(19) Wang, Y.; Chang, W.; Wang, L.; Zhang, Y.; Zhang, Y.; Wang, M.; Wang, Y.; Li, P. A Review of Sources, Multimedia Distribution and Health Risks of Novel Fluorinated Alternatives. *Ecotoxicol. Environ. Saf.* **2019**, *182*, No. 109402.

(20) Eick, S. M.; Goin, D. E.; Trowbridge, J.; Cushing, L.; Smith, S. C.; Park, J.-S.; DeMicco, E.; Padula, A. M.; Woodruff, T. J.; Morello-Frosch, R. Dietary Predictors of Prenatal Per- and Poly-Fluoroalkyl Substances Exposure. *J. Exposure Sci. Environ. Epidemiol.* **2023**, *33* (1), 32−39.

(21) von Stackelberg, K.; Li, M.; Sunderland, E. Results of a National Survey of High-Frequency Fish Consumers in the United States. *Environ. Res.* **2017**, *158*, 126−136.
(22) Wattigney, W. A.; Savadatti, S. S.; Liu, M.; Pavuk, M.; Lewis-Michl, E.; Kannan, K.; Wang, W.; Spliethoff, H.; Marquez-Bravo, L.; Hwang, S.-A. Biomonitoring of Per- and Polyfluoroalkyl Substances in Minority Angler Communities in Central New York State. *Environ. Res.* **2022**, *204*, No. 112309.
(23) Schrenk, D.; Bignami, M.; Bodin, L.; Chipman, J. K.; Mazo, J. del.; Grasl-Kraupp, B.; Hogstrand, C.; Hoogenboom, L. Ron.; Leblanc, J.-C.; Nebbia, C. S.; Nielsen, E.; Ntzani, E.; Petersen, A.; Sand, S.; Vleminckx, C.; Wallace, H.; Barregård, L.; Ceccatelli, S.; Cravedi, J.-P.; Halldorsson, T. I.; Haug, L. S.; Johansson, N.; Knutsen, H. K.; Rose, M.; Roudot, A.-C.; Loveren, H. V.; Vollmer, G.; Mackay, K.; Riolo, F.; Schwerdtle, T. Risk to Human Health Related to the Presence of Perfluoroalkyl Substances in Food. *EFSA J.* **2020**, *18* (9), No. e06223.
(24) Barbo, N.; Stoiber, T.; Naidenko, O. V.; Andrews, D. Q. Locally Caught Freshwater Fish across the United States Are Likely a Significant Source of Exposure to PFOS and Other Perfluorinated Compounds. *Environ. Res.* **2023**, *220*, No. 115165.
(25) Fair, P. A.; Wolf, B.; White, N. D.; Arnott, S. A.; Kannan, K.; Karthikraj, R.; Vena, J. E. Perfluoroalkyl Substances (PFASs) in Edible Fish Species from Charleston Harbor and Tributaries, South Carolina, United States: Exposure and Risk Assessment. *Environ. Res.* **2019**, *171*, 266−277.
(26) MacGillivray, A. R. Temporal Trends of Per- and Polyfluoroalkyl Substances in Delaware River Fish, USA. *Integr. Environ. Assess. Manage.* **2020**, *17* (2), 411−421.
(27) Point, A. D.; Holsen, T. M.; Fernando, S.; Hopke, P. K.; Crimmins, B. S. Trends (2005−2016) of Perfluoroalkyl Acids in Top Predator Fish of the Laurentian Great Lakes. *Sci. Total Environ.* **2021**, *778*, No. 146151.
(28) Pulster, E. L.; Rullo, K.; Gilbert, S.; Ash, T. M.; Goetting, B.; Campbell, K.; Markham, S.; Murawski, S. A. Assessing Per- and Polyfluoroalkyl Substances (PFAS) in Sediments and Fishes in a Large, Urbanized Estuary and the Potential Human Health Implications. *Front. Mar. Sci.* **2022**, *9*, No. 116935, DOI: 10.3389/fmars.2022.1046667.
(29) USEPA. PFAS-Tri National Analysis, 2023. https://www.epa.gov/trinationalanalysis/pfas.
(30) Sedlak, M. D.; Benskin, J. P.; Wong, A.; Grace, R.; Greig, D. J. Per- and Polyfluoroalkyl Substances (PFASs) in San Francisco Bay Wildlife: Temporal Trends, Exposure Pathways, and Notable Presence of Precursor Compounds. *Chemosphere* **2017**, *185*, 1217−1226.
(31) Melwani, A.; Greenfield, B. K.; Yee, D.; Davis, J. A. *Conceptual Foundations for Modeling Bioaccumulation in San Francisco Bay*; San Francisco Estuary Institute: Richmond, CA, 2012.
(32) OEHHA. Eating Fish Safely, 2025. https://oehha.ca.gov/fish/general-health-advice-people-catching-and-eating-sport-fish-california.
(33) USEPA. *Guidance for Assessing Chemical Contaminant Data for Use in Fish Advisories Vol. 1: Fish Sampling and Analysis*, 3rd ed.; United States Environmental Protection Agency: Washington, D.C., 2000.
(34) R Core Team. R: A Language and Environment for Statistical Computing, 2023. https://www.R-project.org/.
(35) Langberg, H. A.; Breedveld, G. D.; Kallenborn, R.; Ali, A. M.; Choyke, S.; McDonough, C. A.; Higgins, C. P.; Jenssen, B. M.; Jartun, M.; Allan, I.; Hamers, T.; Hale, S. E. Human Exposure to Per- and Polyfluoroalkyl Substances (PFAS) via the Consumption of Fish Leads to Exceedance of Safety Thresholds. *Environ. Int.* **2024**, *190*, No. 108844.
(36) EFSA. Perfluorooctane Sulfonate (PFOS), Perfluorooctanoic Acid (PFOA) and Their Salts Scientific Opinion of the Panel on Contaminants in the Food Chain. *EFSA J.* **2008**, *6* (7), 653.
(37) MA Department of Public Health. Evaluation of PFAS in Recreational Waterbodies in Massachusetts, 2021. https://www.mass.gov/doc/technical-basis-for-issuing-fish-advisories-0/download#:~:text=Recent%20surveillance%20of%20surface%20water,)%20(MassDEP%2C%202021)https://www.mass.gov/doc/technical-basis-for-issuing-fish-advisories-0/download#:~:text=Recent%20surveillance%20of%20surface%20water,)%20(MassDEP%2C%202021).
(38) ECCC. *Canadian Environmental Protection Act, 1999 Federal Environmental Quality Guidelines Perfluorooctane Sulfonate (PFOS)*; Environment and Climate Change: Canada, 2018.
(39) Conomos, T. J. Properties and Circulation of San Francisco Bay Waters. In *San Francisco Bay—The Urbanized Estuary*; American Association for the Advancement of Science- Pacific Division, 1979.
(40) Peter, K. T.; Gilbreath, A.; Gonzalez, M.; Tian, Z.; Wong, A.; Yee, D.; Miller, E. L.; Avellaneda, P.; Chen, D.; Patterson, A.; Fitzgerald, N.; Higgins, C. P.; Kolodziej, E. P.; Sutton, R. Storms Mobilize Organophosphate Esters, Bisphenols, PFASs, and Vehicle-Derived Contaminants to San Francisco Bay Watersheds. *Environ. Sci.:Processes Impacts* **2024**, *26*, 1760−1779.
(41) Lin, D.; Méndez, M.; Paterson, K.; Wong, A.; Yee, D.; Sutton, R.; Houtz, E.; Cousins, M.; Fono, L. Residential Wastewater as a Major Source of Per- and Polyfluoroalkyl Substances to Municipal Wastewater. *ACS EST Water* **2024**, *4*, 4847.
(42) Méndez, M.; Trinh, M.; Miller, E.; Lin, D.; Sutton, R. *PFAS in San Francisco Bay Water*; San Francisco Estuary Institute: Richmond, CA, 2022.
(43) Sedlak, M. D.; Sutton, R.; Wong, A.; Lin, D. *Per and Polyfluoroalkyl Substances (PFASs) in San Francisco Bay: Synthesis and Strategy*; San Francisco Estuary Institute: Richmond, CA, 2018; p 130.
(44) Goodrow, S. M.; Ruppel, B.; Lippincott, R. L.; Post, G. B.; Procopio, N. A. Investigation of Levels of Perfluoroalkyl Substances in Surface Water, Sediment and Fish Tissue in New Jersey, USA. *Sci. Total Environ.* **2020**, *729*, No. 138839.
(45) Boatman, A. K.; Chappel, J. R.; Polera, M. E.; Dodds, J. N.; Belcher, S. M.; Baker, E. S. Assessing Per- and Polyfluoroalkyl Substances in Fish Fillet Using Non-Targeted Analyses. *Environ. Sci. Technol.* **2024**, *58* (32), 14486−14495.
(46) Dimitrakopoulou, M.-E.; Karvounis, M.; Marinos, G.; Theodorakopoulou, Z.; Aloizou, E.; Petsangourakis, G.; Papakonstantinou, M.; Stoitsis, G. Comprehensive Analysis of PFAS Presence from Environment to Plate. *npj Sci. Food* **2024**, *8* (1), 80.
(47) Valsecchi, S.; Babut, M.; Mazzoni, M.; Pascariello, S.; Ferrario, C.; De Felice, B.; Bettinetti, R.; Veyrand, B.; Marchand, P.; Polesello, S. Per- and Polyfluoroalkyl Substances (PFAS) in Fish from European Lakes: Current Contamination Status, Sources, and Perspectives for Monitoring. *Environ. Toxicol. Chem.* **2020**, *40* (3), 658−676.
(48) Rüdel, H.; Radermacher, G.; Fliedner, A.; Lohmann, N.; Duffek, A. A Field Study in Support of the Monitoring of Priority Substances in German Freshwater Fish: Derivation of Fillet-to-Whole Fish Conversion Factors. *Environ. Sci. Eur.* **2020**, *32* (1), 13.
(49) Schultes, L.; van Noordenburg, C.; Spaan, K. M.; Plassmann, M. M.; Simon, M.; Roos, A.; Benskin, J. P. High Concentrations of Unidentified Extractable Organofluorine Observed in Blubber from a Greenland Killer Whale (*Orcinus Orca*). *Environ. Sci. Technol. Lett.* **2020**, *7* (12), 909−915.
(50) Lee, K.; Alava, J. J.; Cottrell, P.; Cottrell, L.; Grace, R.; Zysk, I.; Raverty, S. Emerging Contaminants and New POPs (PFAS and HBCDD) in Endangered Southern Resident and Bigg's (Transient) Killer Whales (Orcinus Orca): In Utero Maternal Transfer and Pollution Management Implications. *Environ. Sci. Technol.* **2023**, *57* (1), 360−374.
(51) Barrett, H.; Du, X.; Houde, M.; Lair, S.; Verreault, J.; Peng, H. Suspect and Nontarget Screening Revealed Class-Specific Temporal Trends (2000−2017) of Poly- and Perfluoroalkyl Substances in St. Lawrence Beluga Whales. *Environ. Sci. Technol.* **2021**, *55* (3), 1659−1671.
(52) Wang, Q.; Ruan, Y.; Jin, L.; Zhang, X.; Li, J.; He, Y.; Wei, S.; Lam, J. C. W.; Lam, P. K. S. Target, Nontarget, and Suspect Screening and Temporal Trends of Per- and Polyfluoroalkyl Substances in

Marine Mammals from the South China Sea. *Environ. Sci. Technol.* **2021**, *55* (2), 1045−1056.
(53) Galatius, A.; Bossi, R.; Sonne, C.; Rigét, F. F.; Kinze, C. C.; Lockyer, C.; Teilmann, J.; Dietz, R. PFAS Profiles in Three North Sea Top Predators: Metabolic Differences among Species? *Environ. Sci. Pollut Res.* **2013**, *20* (11), 8013−8020.
(54) Butt, C. M.; Muir, D. C. G.; Mabury, S. A. Biotransformation Pathways of Fluorotelomer-Based Polyfluoroalkyl Substances: A Review. *Environ. Toxicol. Chem.* **2013**, *33* (2), 243−267.
(55) Eriksson, U.; Haglund, P.; Kärrman, A. Contribution of Precursor Compounds to the Release of Per- and Polyfluoroalkyl Substances (PFASs) from Waste Water Treatment Plants (WWTPs). *J. Environ. Sci.* **2017**, *61*, 80−90.
(56) Kaboré, H. A.; Goeury, K.; Desrosiers, M.; Vo Duy, S.; Liu, J.; Cabana, G.; Munoz, G.; Sauvé, S. Novel and Legacy Per- and Polyfluoroalkyl Substances (PFAS) in Freshwater Sporting Fish from Background and Firefighting Foam Impacted Ecosystems in Eastern Canada. *Sci. Total Environ.* **2022**, *816*, No. 151563.
(57) Loi, E. I. H.; Yeung, L. W. Y.; Taniyasu, S.; Lam, P. K. S.; Kannan, K.; Yamashita, N. Trophic Magnification of Poly- and Perfluorinated Compounds in a Subtropical Food Web. *Environ. Sci. Technol.* **2011**, *45* (13), 5506−5513.
(58) Jouanneau, W.; Léandri-Breton, D.-J.; Corbeau, A.; Herzke, D.; Moe, B.; Nikiforov, V. A.; Gabrielsen, G. W.; Chastel, O. A Bad Start in Life? Maternal Transfer of Legacy and Emerging Poly- and Perfluoroalkyl Substances to Eggs in an Arctic Seabird. *Environ. Sci. Technol.* **2022**, *56* (10), 6091−6102.
(59) Kwiatkowski, C. F.; Andrews, D. Q.; Birnbaum, L. S.; Bruton, T. A.; DeWitt, J. C.; Knappe, D. R. U.; Maffini, M. V.; Miller, M. F.; Pelch, K. E.; Reade, A.; Soehl, A.; Trier, X.; Venier, M.; Wagner, C. C.; Wang, Z.; Blum, A. Scientific Basis for Managing PFAS as a Chemical Class. *Environ. Sci. Technol. Lett.* **2020**, *7* (8), 532−543.
(60) European Commission. Directive of the European Parliament and of the Council Amending Directive 2000/60/EC Establishing a Framework for Community Action in the Field of Water Policy, Directive 2006/118/EC on the Protection of Groundwater against Pollution and Deterioration and Directive 2008/105/EC on Environmental Quality Standards in the Field of Water Policy, 2022. https://environment.ec.europa.eu/document/download/6e618dec-c528-4ba8-8900-1e020eefe393_en.
(61) Reinikainen, J.; Bouhoulle, E.; Sorvari, J. Inconsistencies in the EU Regulatory Risk Assessment of PFAS Call for Readjustment. *Environ. Int.* **2024**, *186*, No. 108614.

https://doi.org/10.1021/acsestwater.4c00999
ACS EST Water 2025, 5, 2903−2913